

QICN v21 Internal Note: Independent Finite Separator-Complete Incidence Package and Conditional Closure of I_{int}

QICN internal formal-audit scaffold

2026-05-26

Governance boundary

This document does not certify external support, consciousness, phenomenality, identity transfer, or bridge-burden closure. It is an internal mathematical and implementation scaffold. It does not report human-subject data, external adjudication, reviewer certification, or empirical validation. The package constructed in this note is finite and synthetic; it may be used as a conditional formal witness only after its catalog-completeness premise is independently accepted.

1 Scope and layer separation

Ontology. A system S is treated only as an admissible object with histories, identity data, intervention-response coordinates, and separator tests. No consciousness, agency, subjecthood, or phenomenal identity is attributed.

Mathematical model. The relevant structure is a finite bipartite incidence graph $G_S = (V_\Theta \sqcup V_R, E)$ plus an explicit finite factorization-cut universe.

Implementation. The finite package is represented by `docs/theory/FINITE_SEPARATOR_COMPLETE_PACKAGE_v21.json`; it is audited by `scripts/audit-finite-separator-package.js`. The product-separator negative control is represented separately.

Language. “Separator-complete”, “response-operative”, and “connected” are formal predicates over the finite package. They are not synonyms for consciousness or identity transfer.

Interpretation. The result is aggressive but bounded: accepted finite separator-complete connected incidence implies atomicity relative to the declared finite factorization universe, and therefore conditional closure of I_{int} . It is not a global proof from rigid identity plus continuity plus fidelity alone.

2 Corpus anchors

This note extends the v20 closure note and is anchored to the monolithic Paper 5 claims. Proposition 5.37 is the original integration-transfer anchor, and Theorem 5.11 is the prior conditional theorem under an atomic separator. The formal style follows the framework’s Paper 5 surface [Per26], categorical factorization language [Mac98], and finite graph reasoning compatible with causal graph

vocabulary [Pea09]. These references orient the formalism only; they do not externally validate QICN.

3 Finite package primitives

Definition 3.1 (Finite response-separator incidence package). *A finite response-separator incidence package for a system presentation S is a tuple*

$$\text{Cert}(S) = (V_\Theta, V_R, E, \mathcal{P}, \mathcal{E}, \mathcal{C}),$$

where:

- (i) V_Θ is a finite set of elementary separator vertices declared by a separator catalog $\mathcal{C}$;
- (ii) V_R is a finite set of elementary intervention-response coordinate vertices;
- (iii) $E \subseteq V_\Theta \times V_R$ is the incidence relation;
- (iv) $\mathcal{P}$ is a finite set of typed perturbation records witnessing the edges of E ;
- (v) $\mathcal{E}$ is an enumeration policy for non-trivial binary factorization cuts over $V_\Theta \sqcup V_R$.

An edge $(\theta, r) \in E$ is admissible only if its perturbation record preserves the declared identity typing and history typing while producing a positive separator-margin change.

Definition 3.2 (Independent construction boundary). *The package is independently constructed when V_Θ , V_R , E , and $\mathcal{P}$ are declared before applying the atomicity predicate, and the construction rule does not query whether Θ_S is atomic. Operationally, the v21 package is generated from a literal separator catalog, response-coordinate catalog, and perturbation table. The audit then checks graph connectedness and cut obstruction afterward.*

Definition 3.3 (Finite separator-completeness). *A package $\text{Cert}(S)$ is finite separator-complete relative to its declared universe when:*

- (i) every declared separator in the catalog $\mathcal{C}$ appears exactly as a vertex of V_Θ ;
- (ii) every $\theta \in V_\Theta$ is response-operative, meaning it has at least one incident edge witnessed by a positive typed perturbation record;
- (iii) every $r \in V_R$ has at least one incident edge;
- (iv) every admissible factorization considered by the finite model induces a binary cut listed by the enumeration policy $\mathcal{E}$;
- (v) a factor-local decomposition corresponds to a non-trivial cut with no incidence edge crossing the cut.

This is a finite relative condition. It does not claim completeness over undeclared separators, unrecorded response coordinates, infinite systems, or external empirical systems.

Definition 3.4 (Connected finite incidence). *The incidence graph of $\text{Cert}(S)$ is connected when the undirected bipartite graph $G_S = (V_\Theta \sqcup V_R, E)$ has exactly one connected component after rejecting isolated non-operative vertices.*

Definition 3.5 (Relative atomicity). *The separator family is atomic relative to $\text{Cert}(S)$ when no non-trivial finite factorization cut enumerated by $\mathcal{E}$ is factor-local, i.e. every enumerated non-trivial cut has at least one crossing incidence edge.*

4 The constructed v21 package

Criterion 4.1 (v21 connected finite package). *The file `FINITE_SEPARATOR_COMPLETE_PACKAGE_v21.js on` declares six separator vertices, five response-coordinate vertices, twelve typed incidence edges, and twelve typed perturbation records. The audit enumerates all canonical binary cuts over the eleven vertices, restricts to non-trivial factor cuts containing at least one separator and one response on each side, and verifies that no such cut is edge-closed.*

Criterion 4.2 (Product-separator negative control). *The file `FINITE_SEPARATOR_COMPLETE_PACKAGE_PRODUCT_NEGATIVE_CONTROL_v21.json` deliberately declares two disconnected product components. The same audit must reject it as an atomicity witness. This prevents the validator from merely accepting any syntactically complete finite package.*

5 Main theorem

Theorem 5.1 (Finite separator-complete connected incidence implies relative atomicity). *Let $\text{Cert}(S)$ be a finite separator-complete response-separator incidence package. If its incidence graph is connected, then Θ_S is atomic relative to the finite factorization universe declared by $\text{Cert}(S)$.*

Proof. Assume for contradiction that Θ_S is not atomic relative to $\text{Cert}(S)$. Then there exists an enumerated non-trivial factorization cut

$$V_\Theta \sqcup V_R = A \sqcup B$$

with both A and B containing at least one response-operative separator and at least one response-coordinate vertex, and with no edge of E crossing from A to B .

Because the cut is non-trivial and all represented separators and response vertices are operative, both induced subgraphs are non-empty. The absence of crossing edges means that the graph decomposes as the disjoint union of the subgraph induced by A and the subgraph induced by B . Hence G_S has at least two connected components. This contradicts connectedness. Therefore no enumerated non-trivial factorization cut is factor-local, which is exactly relative atomicity for the finite package. $\square$

Corollary 5.2 (v21 finite-witness conditional closure of I_{int}). *If the v21 finite package is accepted as separator-complete for the declared finite system model, if its audit passes, and if the factorization-loss terms are faithful to history, identity, response, and gauge preservation, then $I_{\text{int}}(S)$ exists and is unique as a gauge-invariant binary factorization-gap invariant relative to that finite model.*

Proof. The audit-passing finite package gives connected finite incidence. The previous theorem gives relative atomicity. The prior conditional theorem under atomic separator, anchored by Theorem 5.11, then applies to the finite model: a zero-loss non-trivial factorization would require a factor-local separator cut, but relative atomicity forbids such a cut. Therefore the factorization-gap predicate defining I_{int} is well-defined and invariant under admissible gauge transformations that preserve the finite package. Uniqueness follows from equality of the binary zero-set for zero-loss factorizations. $\square$

6 Non-circularity and reviewer burden

Remark 6.1 (Reviewer burden). *A reviewer must verify that the separator catalog and response-coordinate catalog can be obtained operationally without first proving atomicity. The admissible*

workflow is: declare elementary separator tests, declare elementary response coordinates, record typed perturbation evidence, construct the incidence graph, enumerate cuts, and only then apply connectedness. The inadmissible workflow is: assume no factor-local separator partition exists and then build the catalog around that assumption. If the inadmissible workflow is used, the v21 certificate is circular and must be rejected.

Remark 6.2 (What this closes). *The v21 construction closes a finite, implementation-checkable route: accepted finite separator-complete connected incidence implies relative atomicity, which gives conditional I_{int} closure. It does not close the global atomic-separator lemma for arbitrary continuous or infinite systems, and it does not transform synthetic internal evidence into empirical validation.*

Remark 6.3 (Negative control interpretation). *The product-separator negative control fails the connectedness and factor-local cut audit. This is the expected behavior. It shows that the validator distinguishes a connected finite witness from a product-decomposed incidence graph. It does not show that real systems satisfy the positive package.*

7 Implementation status

The implementation consists of the package constructor, validator, positive package, negative control package, and generated audit report. The validator is deterministic and computes a package hash. The full result is recorded in `docs/reports/FINITE_SEPARATOR_COMPLETE_PACKAGE_v21_AUDIT.json`. Promotion to a source paper or theorem registry entry still requires human mathematical curation and an explicit decision about the finite-universe premise.

References

- [Mac98] Saunders Mac Lane. *Categories for the Working Mathematician*. 2nd. Vol. 5. Graduate Texts in Mathematics. New York: Springer, 1998.
- [Pea09] Judea Pearl. *Causality: Models, Reasoning, and Inference*. 2nd. Standard reference for causal inference and causal graphs. Cambridge: Cambridge University Press, 2009.
- [Per26] Johnny Andrey Perez Ramirez. “A Structural Criterion for Substrate-Invariant Operational Consciousness”. In: *arXiv preprint* (2026).